

Advanced Econometrics – Exam

June, 2020

Exercise 1 (30%)

Adrian Barnett, Inger Mewburn, and Sara Schorter in their article¹ "Working 9 to 5, not the way to make an academic living: observational analysis of manuscript and peer-review submissions over time" presented a proof how difficult is to be an academician [yes, it is, you can believe us! ;-)].

"The «publish or perish» mantra is well established in academic circles, but do researchers have to sacrifice leisure time to achieve this ideal? [...] The pressure to publish creates a vicious cycle whereby publishing more papers has the potential to push the university up in the rankings, which, in turn, puts more pressure on academics to keep working to maintain their university's place in the rankings (and perhaps their own jobs). At the same time, universities in many countries have seen a massive increase in student numbers and teaching loads that squeeze the available time for research.

Anecdotally at least, academics and medical students are working longer hours to accommodate the demands of research and publishing.

The objective of the study was to determine whether researchers are submitting manuscripts and peer-reviews to BMJ journals out of hours and whether this has changed over time. The data consisted of info on research manuscripts submitted to the British Medical Journal (BMJ) between 2012 and 2019 for which an author's address could be geocoded. The aim of the study is to confirm the hypothesis that considerable amount of submissions takes place on weekends, on national holidays, and out of hours (early mornings and late nights)."²

Your lecturers decided to repeat this analysis with econometric models. First of all, logistic regression was used to estimate the probability of the manuscript was submitted out of hours, that is before 9 a.m. or later than 5 p.m. or on weekends or holidays. The dummy variable *OutOfHours* was equal to one if the submission was made out of hours. In the quality publication table below, you can find parameters' estimates of three logistic regression models. Model (1) contains dummy variables for countries, and where the United States was the base level. Besides the dummies, the model includes *Monday* which is a dummy variable indicating whether the submission was made on Monday (equals to 1) or not. Model (2) is similar to model (1), but it does not contain the *Monday* variable. Last but not least, model (3) is model (2) with insignificant variables removed.

- a) (5%) The variable *Monday* was included in order to verify the hypothesis that even if researchers work on weekends they submit to the BMJ on Monday. How to verify this hypothesis? Explain.
- b) (5%) Which one out of models (1), (2), and (3) should be selected as the most appropriate. Explain.
- c) (5%) "The country someone lives in was the strongest predictor of whether they will work out of hours, suggesting that a «culture of overwork» is a literal thing, not just a figure of speech." Name the country with the highest culture of overwork and one with the lowest "culture of overwork". Explain.
- d) (5%) Interpret the value of the parameter's estimate for Poland in model (3).

¹Barnett Adrian, Mewburn Inger, Schroter Sara. Working 9 to 5, not the way to make an academic living: observational analysis of manuscript and peer-review submissions over time BMJ 2019; 367 :l6460

²Barnett Adrian, Mewburn Inger, Schroter Sara, p. 1.

Table 1:

	<i>Dependent variable:</i>		
	OutOfHours		
	(1)	(2)	(3)
Monday	−0.034		
Australia	−0.009	−0.009	
Austria	0.204	0.206	
Bangladesh	0.570***	0.569***	0.568***
Belgium	−0.003	−0.003	
Brazil	0.360***	0.360***	0.359***
Cameroon	0.678***	0.679***	0.678***
Canada	−0.105**	−0.105**	−0.106***
China	0.790***	0.791***	0.790***
Denmark	−0.414***	−0.414***	−0.415***
Ethiopia	0.061	0.061	
Finland	0.074	0.074	
France	0.322***	0.322***	0.321***
Germany	0.062	0.061	
India	0.488***	0.488***	0.487***
Iran	0.627***	0.628***	0.627***
Ireland	0.081	0.081	
Israel	0.303**	0.304**	0.303**
Italy	0.295***	0.295***	0.294***
Japan	0.589***	0.590***	0.588***
Malaysia	0.373***	0.372***	0.371***
Netherlands	−0.009	−0.009	
New Zealand	−0.218**	−0.218**	−0.219**
Norway	−0.192**	−0.192**	−0.194***
Pakistan	0.565***	0.566***	0.565***
Poland	0.326*	0.325*	0.324*
Portugal	0.655***	0.654***	0.653***
Saudi Arabia	0.480***	0.479***	0.478***
Singapore	0.456***	0.458***	0.457***
South Africa	−0.019	−0.020	
South Korea	0.507***	0.507***	0.506***
Spain	0.244***	0.244***	0.243***
Sweden	0.057	0.057	
Switzerland	0.171**	0.172**	0.170**
Taiwan	0.689***	0.689***	0.687***
Thailand	0.438**	0.439**	0.438**
Turkey	0.637***	0.639***	0.638***
United Kingdom	−0.017	−0.017	
Constant	−0.374***	−0.380***	−0.379***
Bayesian Inf. Crit.	67230.14	67221.27	67109.15
McFadden's R2	0.01843	0.01840	0.01830
Observations	49,464	49,464	49,464
Akaike Inf. Crit.	66,886.590	66,886.530	66,871.310

Note:

*p<0.1; **p<0.05; ***p<0.01

What is more, your lecturers decided to study the number of submissions by hours. For this reason, we aggregated the data over hours for each country. As a result, we received counts, that is numbers of submissions per each hour per each country. Page 4 presents output from a Poisson model for those counts.

e) (5%) Interpret the estimate of the parameter for *hour12* in the Poisson model presented below.

f) (5%) What hour do researchers submit most papers? Interpret the estimate's value.

Call:

```
glm(formula = n ~ as.factor(hour), family = poisson, data = TableOfCounts)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-14.663	-7.160	-2.870	0.008	57.321

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.72759	0.02516	148.169	< 2e-16 ***
as.factor(hour)1	-0.52250	0.04123	-12.672	< 2e-16 ***
as.factor(hour)2	-1.05345	0.04948	-21.292	< 2e-16 ***
as.factor(hour)3	-1.58185	0.06638	-23.830	< 2e-16 ***
as.factor(hour)4	-1.84838	0.07456	-24.791	< 2e-16 ***
as.factor(hour)5	-2.03427	0.07988	-25.468	< 2e-16 ***
as.factor(hour)6	-1.79517	0.07182	-24.996	< 2e-16 ***
as.factor(hour)7	-1.51261	0.06445	-23.469	< 2e-16 ***
as.factor(hour)8	-1.02511	0.04995	-20.522	< 2e-16 ***
as.factor(hour)9	-0.23029	0.03781	-6.090	1.13e-09 ***
as.factor(hour)10	0.38500	0.03261	11.805	< 2e-16 ***
as.factor(hour)11	0.77368	0.03041	25.440	< 2e-16 ***
as.factor(hour)12	0.89944	0.02984	30.143	< 2e-16 ***
as.factor(hour)13	0.82545	0.03017	27.361	< 2e-16 ***
as.factor(hour)14	0.87573	0.02994	29.247	< 2e-16 ***
as.factor(hour)15	1.02804	0.02931	35.070	< 2e-16 ***
as.factor(hour)16	1.04376	0.02925	35.680	< 2e-16 ***
as.factor(hour)17	1.00189	0.02942	34.059	< 2e-16 ***
as.factor(hour)18	0.73011	0.03062	23.841	< 2e-16 ***
as.factor(hour)19	0.39614	0.03254	12.174	< 2e-16 ***
as.factor(hour)20	0.11412	0.03461	3.298	0.000975 ***
as.factor(hour)21	0.17065	0.03415	4.996	5.84e-07 ***
as.factor(hour)22	0.23522	0.03366	6.988	2.79e-12 ***
as.factor(hour)23	0.20938	0.03386	6.184	6.24e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 105214 on 876 degrees of freedom
 Residual deviance: 79954 on 853 degrees of freedom
 AIC: 84146

Number of Fisher Scoring iterations: 6

Exercise 2 (30%)

”So how’d you two meet?” There’s always a story, but the general ways people meet are usually similar. They met through a friend. They met at church. They met through an app.³

How Couples Meet and Stay Together (HCMST) is a study of how Americans meet their spouses and romantic partners.⁴ The study is a nationally representative study of American adults. 3510 adults responded to the survey, 2085 of those was married. In the survey, lots of questions about phone dating apps and other ways of meeting and dating were asked.

Your lecturers decided to analyse the data and confirm some hypotheses. First of all, our analyses were limited to the 5 most popular ways of finding a partner in the United States. The set of alternatives in the model consisted of:

1. Met Online
2. Through Friend
3. Bar or Restaurant
4. Through Co-worker
5. Party

The list of covariates consisted of *Age* - age in years, *Female* - a dummy variable (1 for women, 0 for men), *Degree* - a dummy variable (1 if the person had a degree of any kind: Associate degree, Bachelor’s degree, Master’s degree, Professional or Doctorate degree), and *OppositeSexCouple* - a dummy variable (1 for opposite-sex couples, 0 for the same-sex couples).

Table 2 contains critical values of the χ^2 distribution at the 10% significance level. Table presents marginal effects calculated for mean values of the independent variables.

In this exercise apply 10% significance level.

- a) (5%) Should we estimate the conditional logit model instead of the multinomial logit in this case? Explain.
- b) (5%) How should we interpret R^2 statistic of the model?
- c) (5%) Does the fact of having a degree of any kind impact the way the person finds her/his partner?
- d) (5%) Interpret marginal effects of age on alternative *Online*.
- e) (5%) Interpret marginal effects of opposite-sex couple on alternative *Online*.
- f) (5%) The model was reestimated without variable *female*. The log-likelihood value in the new model was equal to -134.85 . Verify the hypothesis that women find their partner the same way as men do in a formal manner.

Table 2: Critical values of the χ^2 distribution at the 10% significance level

degree of freedom	1	2	3	4	5	6	7	8	9	10
critical value	2.71	4.61	6.25	7.78	9.24	10.64	12.02	13.36	14.68	15.99

³<https://flowingdata.com/2019/03/13/how-people-meet-their-partners/>

⁴<https://data.stanford.edu/hcmst>

Call:

```
mlogit(formula = HowTheyMet ~ 0 | Age + Gender + Degree + OppositeSexCouple,
  data = himym2, reflevel = "MetOnline", shape = "long",
  chid.var = "chid", alt.var = "alt", choice = "HowTheyMet",
  method = "nr")
```

Frequencies of alternatives:

MetOnline	BarOrRestaurant	Party	ThroughCoworker	ThroughFriend
0.340206	0.237113	0.072165	0.185567	0.164948

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
BarOrRestaurant:(intercept)	-1.1708531	0.8604019	-1.3608	0.17357
Party:(intercept)	-2.6043860	1.4588073	-1.7853	0.07422 .
ThroughCoworker:(intercept)	0.1588746	0.7756268	0.2048	0.83770
ThroughFriend:(intercept)	-0.2510572	0.8341606	-0.3010	0.76344
BarOrRestaurant:Age	0.0525126	0.0232790	2.2558	0.02408 *
Party:Age	0.0914868	0.0399539	2.2898	0.02203 *
ThroughCoworker:Age	0.0339905	0.0255251	1.3316	0.18298
ThroughFriend:Age	0.0423283	0.0272390	1.5540	0.12019
BarOrRestaurant:Female	0.1392123	0.6086147	0.2287	0.81907
Party:Female	0.1724922	0.9670042	0.1784	0.85843
ThroughCoworker:Female	-0.2261853	0.6425416	-0.3520	0.72483
ThroughFriend:Female	-1.0312876	0.6743563	-1.5293	0.12619
BarOrRestaurant:Degree	-0.0032208	0.5781834	-0.0056	0.99556
Party:Degree	-1.2972122	0.9460504	-1.3712	0.17032
ThroughCoworker:Degree	-0.3613861	0.6188656	-0.5839	0.55925
ThroughFriend:Degree	0.4561129	0.6856648	0.6652	0.50591
BarOrRestaurant:OppositeSexCouple	-0.5319439	0.7142483	-0.7448	0.45642
Party:OppositeSexCouple	-0.9514622	1.0326843	-0.9213	0.35687
ThroughCoworker:OppositeSexCouple	-1.6255035	0.6851487	-2.3725	0.01767 *
ThroughFriend:OppositeSexCouple	-1.6180822	0.7267227	-2.2265	0.02598 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -133.15

McFadden R²: 0.089487

Likelihood ratio test : chisq = 26.172 (p.value = 0.05165)

Table 3: Marginal effects calculated for mean values of the independent variables

Variable	Alternatives				
	(1)	(2)	(3)	(4)	(5)
Age	-.0078682	.0011193	.0034004	-.0000439	.0033924
Degree	.0153296	.0826683	.0255048	-.0507005	-.0728022
Female	-.0489987	.1285913	-.065626	.0040858	-.0180523
OppositeSexCouple	-.2288816	.1314864	-.0707193	.1598699	.0082445

Exercise 3 (40%)

Dear Students, please meet mink and muskrat.

Figure 1. Muskrat



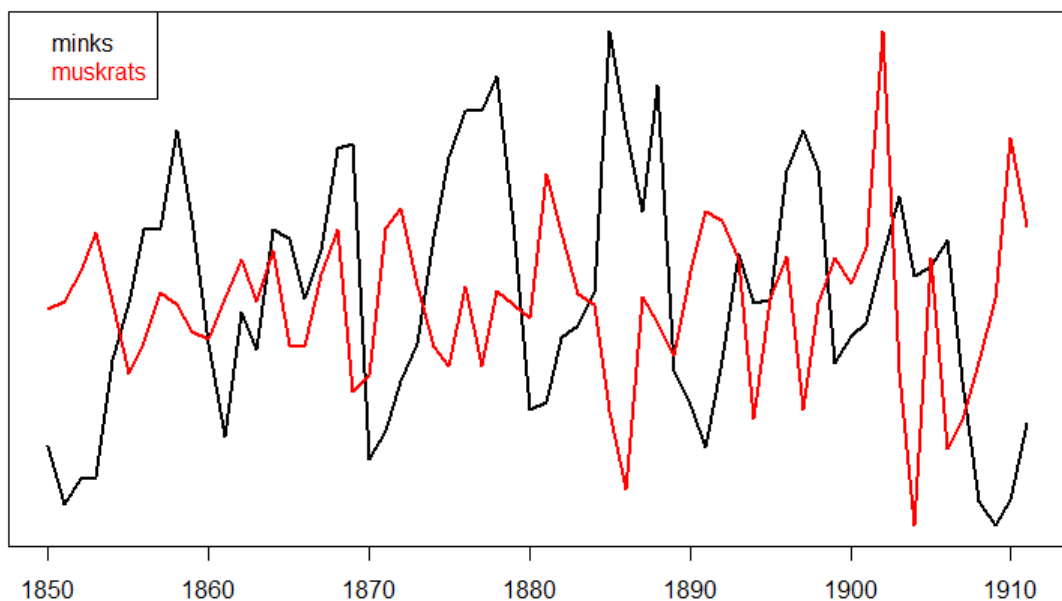
Source: [wikimedia.org/ \(link\)](#)

Figure 2. Mink



Source: [wikimedia.org/ \(link\)](#)

Mink are dark-colored, semiaquatic, carnivorous mammals of the genera *Neovison* and *Mustela*, and part of the family *Mustelidae*, which also includes weasels, otters and ferrets. The American mink's fur has been highly prized for use in clothing, with hunting giving way to farming.⁵ The **muskrat** (*Ondatra zibethicus*), the only species in genus *Ondatra* and tribe *ondatrini*, is a medium-sized semiaquatic rodent native to North America and an introduced species in parts of Europe, Asia, and South America. Muskrats provide an important food resource for many other animals, including **mink**, foxes, coyotes, wolves, lynx, bobcats, bears, eagles, snakes, alligators, and large owls and hawks.⁶



[Let us now discuss] the two series shown in Figure 3. These series record the numbers of mink

⁵<https://en.wikipedia.org/wiki/Mink>

⁶<https://en.wikipedia.org/wiki/Muskrat>

and muskrat furs traded annually by the Hudson's Bay Company in Canada from 1848 to 1909. There is known to be a prey-predator relationship between these two animals and this affects the population dynamics of both of them.⁷

Our goal will be to establish a relationship between number of mink and muskrats. The output below consists of R output obtained from various time series models and analyses.

- a) Is it a case of spurious correlation? Explain.
- b) Let's now inspect the series visually. Are they integrated? Explain.
- c) What is the integration order of `minks` variables? Explain.
- d) What is the integration order of `muskrats` variables? Explain.
- e) Are `minks` and `muskrats` cointegrated? Explain.
- f) Which model should be selected out of the three ARDL models presented below? Explain.
- g) Calculate and interpret the short-term and long-term multipliers for the `dl_3`.
- h) What does the Granger-causality analysis indicate? Explain.

⁷A.C.Harvey, (1989), *Forecasting, Structural Time Series and the Kalman Filter*, Cambridge University Press, p. 9.

```

> testdf2(variable = dt$minks, test.type="nc",
+         max.augmentations = 4, max.order = 3)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -1.0434878 0.2800023             61          0.045580522
2             1 -1.0286117 0.2847290             60          0.005477665
3             2 -0.8920156 0.3281307             59          0.022220094
4             3 -0.8334178 0.3467494             58          0.019830329
5             4 -0.7283168 0.3801440             57          0.225382612
bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1             1.4057657             1.5736972          0.8309401          0.4951558          0.6653684
2             1.3370138             1.4198942          0.9410014          0.5124732          0.7008781
3             0.1392349             0.3560613          0.8815031          0.9327506          0.9491619
4             0.1761922             0.2301956          0.8880120          0.9156729          0.9725735
5             0.7724574             1.2685787          0.6349689          0.6796151          0.7366075

> testdf2(variable = dt$muskrats, test.type="nc",
+         max.augmentations = 4, max.order = 3)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -6.323482  0.01             61          0.13383622
2             1 -6.219694  0.01             60          0.05401426
3             2 -4.081416  0.01             59          0.08147088
4             3 -5.258275  0.01             58          0.19525076
5             4 -4.903783  0.01             57          0.22875325
bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1             4.9190242          6.0195304          0.7144872          0.08547665          0.1106639
2             0.9545065          2.0253250          0.8162198          0.62048536          0.5671671
3             0.4988606          0.7649031          0.7753139          0.77924459          0.8578420
4             0.9183355          1.2441547          0.6585819          0.63180924          0.7424348
5             1.9146262          2.0947157          0.6324498          0.38392307          0.5529826

> testdf2(variable = diff(dt$minks), test.type="nc",
+         max.augmentations = 4, max.order = 3)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -7.464321  0.01             60          0.002562517
2             1 -6.093589  0.01             59          0.003487338
3             2 -4.906297  0.01             58          0.005505509
4             3 -5.035137  0.01             57          0.196935050
5             4 -5.147511  0.01             56          0.412764193
bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1             1.34064025          1.4332867          0.9596273          0.5115448          0.6977510
2             0.08212202          0.3520402          0.9529093          0.9597706          0.9499615
3             0.13588402          0.1852202          0.9408520          0.9343147          0.9799392
4             0.73381946          1.2342153          0.6572062          0.6928722          0.7448103
5             1.58478627          2.7552686          0.5205698          0.4527600          0.4309165

```

```
> testdf2(variable = diff(dt$muskrats), test.type="nc",
+         max.augmentations = 4, max.order = 3)
      augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 -10.042191  0.01           60           0.98382055
2           1 -10.171713  0.01           59           0.01475214
3           2  -5.690695  0.01           58           0.01099714
4           3  -5.463414  0.01           57           0.01866789
5           4  -4.328418  0.01           56           0.01579248
      bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1           12.9114283      13.3956031      0.3212574      0.001571517      0.003854706
2           0.6313132      0.6958729      0.9033280      0.729309844      0.874174246
3           0.5801917      0.7266606      0.9164812      0.748191857      0.866915912
4           0.1421436      0.5131459      0.8913229      0.931395027      0.915993965
5           0.2775689      0.6409912      0.8999946      0.870415629      0.886987529

> model.coint = lm(minks~muskrats, data=dt)
> summary(model.coint)
```

Call:

```
lm(formula = minks ~ muskrats, data = dt)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-34243 -13125  -1458   11450   33489
```

Coefficients:

```
              Estimate Std. Error t value      Pr(>|t|)
(Intercept) 52091.16976  2211.27859   23.557 <0.0000000000000002 ***
muskrats     -0.02730    0.01069   -2.554      0.0132 *
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 17380 on 60 degrees of freedom

Multiple R-squared: 0.09809, Adjusted R-squared: 0.08306

F-statistic: 6.525 on 1 and 60 DF, p-value: 0.01319

```
> testdf2(variable = resid(model.coint), test.type="nc",
+         max.augmentations = 4, max.order = 3)
      augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 -4.208015  0.01           61           0.243551850
2           1 -4.465550  0.01           60           0.014795433
3           2 -3.794224  0.01           59           0.009270986
4           3 -4.094188  0.01           58           0.072743384
5           4 -3.870033  0.01           57           0.086777440
      bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3
1           1.4452028      1.4598256      0.6216525      0.4854877      0.6915731
2           0.2958177      0.4770259      0.9031869      0.8625097      0.9239089
3           0.1648981      0.2964598      0.9232935      0.9208584      0.9606929
4           0.2411588      0.2817348      0.7873835      0.8864067      0.9634263
5           0.2214667      0.2256841      0.7683147      0.8951774      0.9733402
```

```
> dl_1 <- dynlm(minks~muskrats+L(muskrats, c(1:4)), data = dt)
> summary(dl_1)
```

Time series regression with "ts" data:
Start = 5, End = 62

Call:
dynlm(formula = minks ~ muskrats + L(muskrats, c(1:4)), data = dt)

Residuals:

Min	1Q	Median	3Q	Max
-34308	-9759	-3176	11895	36776

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	53569.697645	2103.593494	25.466	<0.00000000000000002 ***
muskrats	-0.020417	0.011362	-1.797	0.0781 .
L(muskrats, c(1:4))1	0.006498	0.011067	0.587	0.5597
L(muskrats, c(1:4))2	-0.001450	0.012029	-0.121	0.9045
L(muskrats, c(1:4))3	0.010489	0.011409	0.919	0.3622
L(muskrats, c(1:4))4	0.026186	0.011759	2.227	0.0303 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15990 on 52 degrees of freedom
Multiple R-squared: 0.2197, Adjusted R-squared: 0.1447
F-statistic: 2.929 on 5 and 52 DF, p-value: 0.02103

```
> bgtest(residuals(dl_1)~1, order = 1)
Breusch-Godfrey test for serial correlation of order up to 1
data: residuals(dl_1) ~ 1
LM test = 11.214, df = 1, p-value = 0.0008118
```

```
> bgtest(residuals(dl_1)~1, order = 2)
Breusch-Godfrey test for serial correlation of order up to 2
data: residuals(dl_1) ~ 1
LM test = 12.792, df = 2, p-value = 0.001668
```

```
> bgtest(residuals(dl_1)~1, order = 3)
Breusch-Godfrey test for serial correlation of order up to 3
data: residuals(dl_1) ~ 1
LM test = 13.107, df = 3, p-value = 0.004411
```

```
> bgtest(residuals(dl_1)~1, order = 4)
Breusch-Godfrey test for serial correlation of order up to 4
data: residuals(dl_1) ~ 1
LM test = 13.336, df = 4, p-value = 0.009747
```

```
> dl_2 <- dynlm(minks~muskrats+L(muskrats, c(1:4))+L(minks, c(1:4)), data = dt)
> summary(dl_2)
```

Time series regression with "ts" data:

Start = 5, End = 62

Call:

```
dynlm(formula = minks ~ muskrats + L(muskrats, c(1:4)) + L(minks,
  c(1:4)), data = dt)
```

Residuals:

Min	1Q	Median	3Q	Max
-31397.6	-6802.6	-932.9	6022.6	29495.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	34042.9801824	8375.0902798	4.065	0.000177 ***
muskrats	0.0156973	0.0101512	1.546	0.128589
L(muskrats, c(1:4))1	0.0174911	0.0093296	1.875	0.066913 .
L(muskrats, c(1:4))2	-0.0042435	0.0103725	-0.409	0.684275
L(muskrats, c(1:4))3	0.0007181	0.0099668	0.072	0.942861
L(muskrats, c(1:4))4	0.0214556	0.0090260	2.377	0.021484 *
L(minks, c(1:4))1	0.7414231	0.1404228	5.280	0.00000309 ***
L(minks, c(1:4))2	-0.0128046	0.1746147	-0.073	0.941848
L(minks, c(1:4))3	-0.1200762	0.1722362	-0.697	0.489066
L(minks, c(1:4))4	-0.2480640	0.1562505	-1.588	0.118942

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11880 on 48 degrees of freedom

Multiple R-squared: 0.6028, Adjusted R-squared: 0.5283

F-statistic: 8.092 on 9 and 48 DF, p-value: 0.0000003539

```
> bgtest(residuals(dl_2)~1, order = 1)
```

Breusch-Godfrey test for serial correlation of order up to 1

data: residuals(dl_2) ~ 1

LM test = 1.6827, df = 1, p-value = 0.1946

```
> bgtest(residuals(dl_2)~1, order = 2)
```

Breusch-Godfrey test for serial correlation of order up to 2

data: residuals(dl_2) ~ 1

LM test = 3.6851, df = 2, p-value = 0.1584

```
> bgtest(residuals(dl_2)~1, order = 3)
```

Breusch-Godfrey test for serial correlation of order up to 3

data: residuals(dl_2) ~ 1

LM test = 3.9002, df = 3, p-value = 0.2724

```
> bgtest(residuals(dl_2)~1, order = 4)
```

Breusch-Godfrey test for serial correlation of order up to 4

data: residuals(dl_2) ~ 1

LM test = 3.9854, df = 4, p-value = 0.408

```
> dl_3 <- dynlm(minks~muskrats+L(muskrats, c(1:4))+L(minks, 1), data = dt)
> summary(dl_3)
```

Time series regression with "ts" data:

Start = 5, End = 62

Call:

```
dynlm(formula = minks ~ muskrats + L(muskrats, c(1:4)) + L(minks,
  1), data = dt)
```

Residuals:

Min	1Q	Median	3Q	Max
-33192	-6658	33	8503	28518

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17350.5703394	6448.6537264	2.691	0.00962 **
muskrats	0.0061063	0.0100024	0.610	0.54425
L(muskrats, c(1:4))1	0.0190468	0.0089342	2.132	0.03786 *
L(muskrats, c(1:4))2	-0.0002385	0.0094248	-0.025	0.97991
L(muskrats, c(1:4))3	0.0128273	0.0089459	1.434	0.15771
L(muskrats, c(1:4))4	0.0245691	0.0092152	2.666	0.01025 *
L(minks, 1)	0.6718956	0.1156568	5.809	0.000000407 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12530 on 51 degrees of freedom

Multiple R-squared: 0.5305, Adjusted R-squared: 0.4752

F-statistic: 9.603 on 6 and 51 DF, p-value: 0.0000004651

```
> bgtest(residuals(dl_3)~1, order = 1)
```

Breusch-Godfrey test for serial correlation of order up to 1

data: residuals(dl_3) ~ 1

LM test = 0.00096322, df = 1, p-value = 0.9752

```
> bgtest(residuals(dl_3)~1, order = 2)
```

Breusch-Godfrey test for serial correlation of order up to 2

data: residuals(dl_3) ~ 1

LM test = 0.35596, df = 2, p-value = 0.837

```
> bgtest(residuals(dl_3)~1, order = 3)
```

Breusch-Godfrey test for serial correlation of order up to 3

data: residuals(dl_3) ~ 1

LM test = 0.68608, df = 3, p-value = 0.8765

```
> bgtest(residuals(dl_3)~1, order = 4)
```

Breusch-Godfrey test for serial correlation of order up to 4

data: residuals(dl_3) ~ 1

LM test = 1.7035, df = 4, p-value = 0.7901


```
> grangertest(x=mm$mink, y=mm$muskrat, order=4)
```

Granger causality test

Model 1: mm\$muskrat ~ Lags(mm\$muskrat, 1:4) + Lags(mm\$mink, 1:4)

Model 2: mm\$muskrat ~ Lags(mm\$muskrat, 1:4)

	Res.Df	Df	F	Pr(>F)
1		49		
2	53	-4	6.6584	0.0002316 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> grangertest(x=mm$muskrat, y=mm$mink, order=4)
```

Granger causality test

Model 1: mm\$mink ~ Lags(mm\$mink, 1:4) + Lags(mm\$muskrat, 1:4)

Model 2: mm\$mink ~ Lags(mm\$mink, 1:4)

	Res.Df	Df	F	Pr(>F)
1		49		
2	53	-4	1.934	0.1196

